Layer 12: SWV Peak Extraction and Signal Processing

From Raw Voltammogram to Peak Areas with Quantified Uncertainty

Eton iGEM 2026, Dry Lab | Layer 12 of 17-Layer Biosensor Pipeline | `peak_extraction.py`

Layer 11 produces a theoretical net SWV current trace $\Delta I(E)$. The instrument measures a noisy, baseline-corrupted version of this trace. Layer 12 takes the raw measured array $\{(E_n, \Delta I_n)\}$ and extracts two numbers: the MB peak area $A_{\text{area}}^{\text{MB}}$ and the Fc peak area $A_{\text{area}}^{\text{Fc}}$, each with a propagated uncertainty σ. These two quantities are the sole inputs to Layer 13.

The importance of this layer cannot be overstated. No matter how accurate the physical models in Layers 0–11 are, a systematic error introduced here (wrong baseline, clipped peak, noise spike misidentified as signal) corrupts every downstream calculation: the ratiometric signal F_k, the apparent rate κ_k, and the final enzyme concentration. The pipeline therefore applies four sequential operations: noise filtering (Section II), background subtraction (Section III), peak fitting (Section IV), and quality diagnostics (Section VI).

I. Physical Shape of the SWV Peak

Before fitting a model to the data, it is essential to derive the expected peak shape from first principles so that the choice of fitting function can be justified rather than assumed.

Fermi-Dirac occupancy and the sech^2 peak

From Layer 11, the net SWV current at potential step n is $\Delta I_n = I_{\text{fwd}}(E_n + E_{\text{sw}}) - I_{\text{rev}}(E_n - E_{\text{sw}})$. In the reversible limit ($\kappa_{\text{SW}} \gg 1$), each half-cycle reaches Nernst equilibrium before current is sampled. The equilibrium surface coverage of oxidised MB at potential E is set by the Fermi-Dirac occupancy function:

$$\Gamma_{\text{ox}}^{\infty}(E) = \frac{\Gamma_{\text{intact}}}{1 + \exp\!\big(n_e F(E - E_{\text{MB}}^{\circ\prime})/RT\big)} \tag{I.1}$$

This equation states that at potentials much more positive than $E_{\text{MB}}^{\circ\prime}$, MB is predominantly oxidised ($\Gamma_{\text{ox}} \to \Gamma_{\text{intact}}$); at potentials much more negative, it is predominantly reduced ($\Gamma_{\text{ox}} \to 0$). The transition is centred at $E = E_{\text{MB}}^{\circ\prime}$ and its sharpness is set by $RT/(n_e F)$: lower temperature or more electrons transferred gives a sharper transition.

The net SWV current is proportional to the difference in equilibrium coverage between the two half-cycle potentials:

$$\Delta I(E) \propto \Gamma_{\text{ox}}^{\infty}(E + E_{\text{sw}}) - \Gamma_{\text{ox}}^{\infty}(E - E_{\text{sw}}) \tag{I.2}$$

For small E_{sw} relative to $RT/(n_e F)$, this difference approximates the derivative of $\Gamma_{\text{ox}}^{\infty}$ with respect to E, multiplied by $2E_{\text{sw}}$. Computing that derivative from Eq. (I.1):

$$\frac{d\Gamma_{\text{ox}}^{\infty}}{dE} = -\frac{n_e F \Gamma_{\text{intact}}}{4RT} \operatorname{sech}^2\!\left(\frac{n_e F(E - E_{\text{MB}}^{\circ\prime})}{2RT}\right) \tag{I.3}$$

The net SWV current therefore has the shape:

$$\boxed{\Delta I(E) \propto \Gamma_{\text{intact}} \cdot \operatorname{sech}^2\!\left(\frac{n_e F(E - E_{\text{MB}}^{\circ\prime})}{2RT}\right)} \tag{I.4}$$

This result, confirmed by Mirceski et al. [1] for surface-confined reversible systems, tells us three things directly. First, the peak is *proportional to* Γ_{intact}: it is a direct measure of the remaining intact probe density after cleavage, which is the quantity Layer 13 must extract. Second, the peak is centred exactly at $E_{\text{MB}}^{\circ\prime}$, allowing a cross-check against the pH-corrected value from Layer 10. Third, the peak shape is entirely determined by the thermodynamic ratio $RT/(n_e F)$: it is not a free parameter of the fitting model.

Full-width at half-maximum and the Gaussian approximation

The FWHM of Eq. (I.4) is obtained by solving $\operatorname{sech}^2(x) = 1/2$:

$$\text{FWHM} = \frac{2 \times 2\ln(1+\sqrt{2}) \times RT}{n_e F} = \frac{2 \times 1.763 \times RT}{n_e F} = \frac{3.526 \times 8.314 \times 310}{2 \times 96485} \approx 47 \text{ mV} \tag{I.5}$$

at $T = 310$ K, $n_e = 2$. The corresponding Gaussian standard deviation is:

$$\sigma_E^{\text{pred}} = \frac{\text{FWHM}}{2\sqrt{2\ln 2}} = \frac{\text{FWHM}}{2.355} \approx 20 \text{ mV} \tag{I.6}$$

The Gaussian function $\exp(-x^2/2)$ approximates $\text{sech}^2(ax)$ to within 5% for $|x| \leq 1.5\sigma$ when the Gaussian σ is matched to the sech^2 FWHM via Eq. (I.6). Beyond $\pm 1.5\sigma$, the sech^2 has heavier tails than the Gaussian; this 5% discrepancy in the wings contributes approximately 0.3% error to the integrated peak area. The Gaussian fit is therefore an excellent approximation for area extraction and is preferred over sech^2 for numerical stability.

In the quasi-reversible regime ($\kappa_{\text{SW}} \sim 1$), the peak broadens and its height decreases compared to the reversible limit. Mirceski et al. [1] show that the peak shape remains approximately Gaussian in the quasi-reversible regime, with σ_E increasing beyond the reversible prediction. A measured $\sigma_E > 1.5 \times \sigma_E^{\text{pred}}$ is therefore a sign that the system is in the quasi-reversible regime at the chosen f_{SWV}, not of heterogeneous SAM chemistry.

II. Savitzky-Golay Noise Filtering

The convolution equation

The potentiostat samples the current at $N \approx 650$ discrete potential steps (for a -0.65 to $+0.65$ V scan at 2 mV step, from Baker 2006 [3] and Lubin 2010 [4]). Each sampled value ΔI_n carries per-point noise δ_n with standard deviation σ_{noise} (from the transimpedance amplifier and analogue-to-digital converter, Layer 14). The Savitzky-Golay (SG) filter [2] replaces each point with a weighted sum of its neighbours:

$$\hat{I}_n = \sum_{k=-m}^{m} c_k \, \Delta I_{n+k} \tag{II.1}$$

where m is the half-window width and the weights $\{c_k\}$ are determined by fitting a polynomial of degree p to the $2m+1$ points in the window and reading off the polynomial value at the centre point. The key property of these weights is that they satisfy:

$$\sum_{k=-m}^{m} c_k = 1, \qquad \sum_{k=-m}^{m} c_k \, k^j = 0 \quad \text{for } j = 1, \ldots, p \tag{II.2}$$

The first condition in Eq. (II.2) ensures that a constant signal passes through the filter unchanged (no DC bias). The second condition ensures that polynomial signals of degree up to p also pass unchanged, which is why the filter preserves curved features such as peak tops.

Why SG outperforms a running average for peaked data

A running average (boxcar filter) is the special case $c_k = 1/(2m+1)$ for all k, corresponding to $p = 0$ (degree-0 polynomial, a flat line). A flat-line fit is a poor model for a peak: it systematically underestimates the signal at the peak apex and overestimates it on the flanks, broadening the peak in proportion to the window width. The SG filter with $p = 3$ fits a cubic polynomial locally, which can reproduce the curvature at the peak top without distorting it, as long as the peak is wider than the window.

Noise reduction factor

Because the weights $\{c_k\}$ are fixed (not random), and because the noise on each point δ_n is uncorrelated (white noise from the ADC and TIA), the noise on the smoothed point $\hat{I}_n$ is:

$$\sigma_{\hat{I}} = \sigma_{\text{noise}} \sqrt{\sum_k c_k^2} \tag{II.3}$$

For the symmetric SG filter with $m = 5$ (11 points), $p = 3$, the tabulated sum $\sum_k c_k^2 \approx 0.233$ [2], giving:

$$\sigma_{\hat{I}} = \sigma_{\text{noise}} \times \sqrt{0.233} \approx 0.48 \, \sigma_{\text{noise}} \tag{II.4}$$

The filter reduces per-point noise by a factor of $\approx 2.1\times$ relative to the unsmoothed data. This is meaningful: the per-point SNR on the MB peak is approximately $0.27\,\mu\text{A}/\sigma_{\text{noise}}$; doubling the per-point SNR halves the uncertainty on the fitted peak amplitude.

Parameter choice

The window half-width $m = 5$ (11 points, 22 mV) is chosen to be well below the peak FWHM of 47 mV (Eq. I.5), ensuring no significant peak broadening. The polynomial degree $p = 3$ is the minimum needed to capture the curvature at the peak apex (a quadratic feature); degree 5 would give marginal improvement in peak preservation at the cost of reduced noise suppression. These choices are consistent with the recommendations of Savitzky and Golay [2] for spectroscopic peaks with similar width-to-window ratios.

III. Baseline Subtraction

Origin of the baseline

Even after SWV differential cancellation (Layer 11, Section VII), a non-zero slowly varying baseline remains in the measured trace for three reasons. First, the capacitive transient is not perfectly symmetric between forward and reverse half-cycles when the potential step ΔE_s is non-zero, leaving a residual proportional to $dC_{\mathrm{dl}}/dE \times \Delta E_s$. Second, the O_2 reduction wave contributes a small monotone cathodic current across the scan (Layer 11, Section VIII). Third, unresolved faradaic contributions from adventitious species in the sample matrix may add broad features.

The baseline is not a physical signal of interest; it is an additive artefact that, if not removed, biases the peak area. A 1 nA baseline elevation across a 200 mV peak window adds approximately 1 nA $\times$ 0.2 V = 0.2 nA $\cdot$ V to the fitted area, comparable to the area of the MB peak itself at low probe coverage.

Peak windows and anchor regions

Two *peak windows* are defined as the voltage ranges expected to contain Faradaic current. Everything outside these windows is treated as baseline and used to anchor the baseline fit.

The window boundaries are set at $\pm 3\sigma_E^{\mathrm{pred}}$ from the expected peak centres:

$$\text{MB window:} \quad [E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH}) - 3\sigma_E^{\mathrm{pred}},\ E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH}) + 3\sigma_E^{\mathrm{pred}}] = [E_{\mathrm{MB}}^{\circ\prime} - 60\ \mathrm{mV},\ E_{\mathrm{MB}}^{\circ\prime} + 60\ \mathrm{mV}] \tag{III.1}$$

$$\text{Fc window:} \quad [E_{\mathrm{Fc}}^{\circ\prime} - 3\sigma_E^{\mathrm{pred,Fc}},\ E_{\mathrm{Fc}}^{\circ\prime} + 3\sigma_E^{\mathrm{pred,Fc}}] \approx [E_{\mathrm{Fc}}^{\circ\prime} - 90\ \mathrm{mV},\ E_{\mathrm{Fc}}^{\circ\prime} + 90\ \mathrm{mV}] \tag{III.2}$$

The 3σ guard band ensures that less than 0.3% of the Gaussian peak area lies outside the window and therefore the anchor points carry essentially no Faradaic signal. The Fc window is wider (90 mV vs 60 mV) because the Fc peak, with $n_e = 1$, has FWHM $\approx$ 94 mV at 37°C, giving $\sigma_E^{\mathrm{pred,Fc}} \approx 40$ mV.

Baseline model and fit

A cubic polynomial is fitted by least squares to the N_{base} points in the anchor regions. The polynomial is then evaluated across the full scan range and subtracted from the smoothed data. The baseline-subtracted current at step n is:

$$\Delta I_n^{\mathrm{corr}} = \hat{I}_n - \left(b_0 + b_1 E_n + b_2 E_n^2 + b_3 E_n^3\right) \tag{III.3}$$

The cubic polynomial captures the slowly varying capacitive slope and any mild nonlinearity; a linear fit would suffice for a flat baseline but fails if the O_2 wave tail introduces curvature at the cathodic end of the scan.

Quality check on baseline fit

The RMS residual of the polynomial fit over the anchor points is compared to the expected noise level σ_{noise}:

$$Q_{\mathrm{base}} = \frac{1}{N_{\mathrm{base}}} \sum_{n \in \mathrm{anchor}} \left(\hat{I}_n - \hat{b}(E_n)\right)^2 / \sigma_{\mathrm{noise}}^2 \tag{III.4}$$

Q_{base} follows a χ^2 distribution with $N_{\mathrm{base}} - 4$ degrees of freedom under the null hypothesis that the baseline is truly cubic. If $Q_{\mathrm{base}} > 5$, the baseline is non-polynomial, indicating possible electrode fouling, an unresolved interfering wave, or a bubble at the electrode surface. The scan is flagged and excluded from averaging.

IV. Gaussian Peak Fitting and Area Extraction

The Gaussian model

After baseline subtraction, the baseline-corrected current in each peak window is fitted to a Gaussian:

$$\Delta I^{\mathrm{corr}}(E) = A_{\mathrm{peak}} \exp\!\left(-\frac{(E - E_{\mathrm{peak}})^2}{2\sigma_E^2}\right) \tag{IV.1}$$

This model has three free parameters. A_{peak} (nA) is the peak amplitude: it is proportional to Γ_{intact} (the quantity Layer 13 needs) but also to $f(\kappa_{\mathrm{SW}})$, the kinetic factor from Layer 11 that depends on k_{ET}^0 and the scan frequency f_{SWV}. Because k_{ET}^0 varies between electrodes (different probe lengths, different tether conformations), A_{peak} alone is not a reliable measure of Γ_{intact}. E_{peak} (V) is the measured peak centre: it should agree with the Layer 10 prediction $E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH})$ within a tolerance set by the on-chip reference electrode drift (typically $\pm$25 mV; Baker 2006 [3], Lubin 2010 [4]). A shift outside this tolerance indicates reference electrode failure, sample pH far from expectation, or a misidentified peak. σ_E (V) is the peak width: its predicted value is $\sigma_E^{\mathrm{pred}} \approx 20$ mV (Eq. I.6); deviations carry diagnostic information (see Section VI).

Why peak area rather than peak height

The integrated area under the Gaussian is:

$$A_{\text{area}} = \int_{-\infty}^{+\infty} A_{\text{peak}} \exp\!\left(-\frac{(E - E_{\text{peak}})^2}{2\sigma_E^2}\right) dE = A_{\text{peak}}\, \sigma_E\, \sqrt{2\pi} \tag{IV.2}$$

This equation says that the area is the product of the peak height A_{peak} and the width σ_E, scaled by $\sqrt{2\pi}$. The key physical point is that as the system moves from the reversible to the quasi-reversible regime (lower f_{SWV}, slower k_{ET}^0), the peak becomes shorter and broader simultaneously: A_{peak} decreases while σ_E increases, with the product $A_{\text{peak}} \times \sigma_E$ approximately conserved. This charge conservation follows from the fact that the total charge passed during a sweep through the peak is $Q = n_e F A_{\text{true}} \Gamma_{\text{intact}}$, which depends only on the physical state of the electrode (probe coverage and electrode area) and not on the scan kinetics.

The amplitude A_{peak} therefore conflates the kinetic regime with the surface coverage, whereas the area A_{area} is approximately proportional to Γ_{intact} alone. White and Plaxco [5] confirm experimentally that the SWV signal gain (effectively the peak area) is the more robust metric for quantifying surface coverage across a range of f_{SWV} values. We therefore use A_{area} as the primary output of this layer, not A_{peak}.

Nonlinear least-squares fitting and the covariance matrix

The three parameters $(A_{\text{peak}}, E_{\text{peak}}, \sigma_E)$ are determined by minimising the sum of squared residuals between the data and Eq. (IV.1). The fitting algorithm (Levenberg-Marquardt, as implemented in `scipy.optimize.curve_fit`) provides, in addition to the best-fit parameters, the 3×3 covariance matrix Σ. The diagonal elements give the variances: $\sigma^2_{A_{\text{peak}}} = \Sigma_{11}$, $\sigma^2_{E_{\text{peak}}} = \Sigma_{22}$, $\sigma^2_{\sigma_E} = \Sigma_{33}$. The off-diagonal element Σ_{13} gives the covariance between A_{peak} and σ_E.

This covariance is physically important and must not be ignored. When a peak is slightly wider than the model predicts, the least-squares algorithm responds by reducing A_{peak} and increasing σ_E, keeping their product approximately constant. The amplitude and width are therefore negatively correlated: $\Sigma_{13} < 0$. Propagating uncertainty in $A_{\text{area}} = A_{\text{peak}}\, \sigma_E\, \sqrt{2\pi}$ while ignoring this covariance overestimates $\sigma_{A_{\text{area}}}$. The correct propagation is:

$$\sigma^2_{A_{\text{area}}} = 2\pi \Big[\sigma_E^2\, \sigma^2_{A_{\text{peak}}} + A^2_{\text{peak}}\, \sigma^2_{\sigma_E} + 2\, A_{\text{peak}}\, \sigma_E\, \Sigma_{13}\Big] \tag{IV.3}$$

The third term (the covariance contribution) is negative, so ignoring it gives a conservative (upper-bound) estimate of $\sigma_{A_{\text{area}}}$.

Initial guesses and fitting bounds

The fitting algorithm requires initial guesses for the three parameters. These are obtained from the smoothed, baseline-subtracted data: the initial A_{peak} is the maximum value within the peak window, E_{peak} is the potential at that maximum, and σ_E is set to the predicted value σ_E^{pred}. Bounds are imposed: $A_{\text{peak}} \in [0, 2\max(\Delta I^{\text{corr}})]$, $E_{\text{peak}} \in [E^{\circ\prime}_{\text{MB}}(\text{pH}) \pm 75\ \text{mV}]$, $\sigma_E \in [0.5\sigma_E^{\text{pred}}, 5\sigma_E^{\text{pred}}]$. These bounds prevent the algorithm from converging to unphysical solutions (negative amplitude, unreasonably sharp or broad peaks).

V. Two-Gaussian Simultaneous Fit

When both MB and Fc peaks are extracted from the same scan, they are fitted simultaneously with a shared baseline model:

$$\Delta I(E) = A^{\text{MB}}_{\text{peak}}\, G_{\text{MB}}(E) + A^{\text{Fc}}_{\text{peak}}\, G_{\text{Fc}}(E) + b_0 + b_1 E \tag{V.1}$$

where $G_j(E) = \exp(-(E - E^j_{\text{peak}})^2/(2\sigma_E^{j\,2}))$. The simultaneous fit prevents the following bias: if the two peaks are fitted sequentially and the first fit has slightly wrong wings, the residual from the first peak is absorbed into the baseline of the second, systematically biasing the second area.

The two peaks are resolvable as independent Gaussians when their separation exceeds $3(\sigma_E^{\text{MB}} + \sigma_E^{\text{Fc}})$. For $\sigma_E^{\text{MB}} \approx 20$ mV and $\sigma_E^{\text{Fc}} \approx 40$ mV, the resolvability criterion requires $|E^{\text{Fc}}_{\text{peak}} - E^{\text{MB}}_{\text{peak}}| > 180$ mV. The expected separation is approximately 690 mV (Layer 10, Section VIII), comfortably exceeding this threshold. The two-Gaussian model is therefore never degenerate for this system.

VI. Quality Diagnostics

Every extracted peak is subjected to the following automated diagnostics. Each diagnostic flag has a specific physical interpretation.

Diagnostic	Flag condition	Expected value	Physical interpretation
Peak width σ_E^{MB}	$> 2\sigma_E^{\mathrm{pred}}$	20 mV	SAM is heterogeneous (two populations of probe with different $E_{\mathrm{MB}}^{\circ\prime}$), or dissolved MB is contributing a broader diffusion-controlled peak
Peak width σ_E^{MB}	$< 0.5\sigma_E^{\mathrm{pred}}$	20 mV	Noise spike misidentified as peak; or the peak is not a Faradaic signal
Peak position $\lvert E_{\mathrm{peak}}^{\mathrm{MB}} - E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH})\rvert$	> 25 mV	< 5 mV	Reference electrode drift, sample pH far from the assumed value, or a different redox-active species is being fitted
Fit χ^2/ν	> 5	≈ 1	Non-Gaussian peak shape; the sech^2 vs Gaussian discrepancy is only 5% at the wings, so a value > 5 indicates a real non-Gaussianity (asymmetric peak from two probe populations, or double-MB peak)
Fc amplitude stability $\lvert A_{\mathrm{peak}}^{\mathrm{Fc}}(i) - \bar{A}_{\mathrm{peak}}^{\mathrm{Fc}}\rvert / \bar{A}_{\mathrm{peak}}^{\mathrm{Fc}}$	$> 10\%$ between scans	$< 2\%$	Fc SAM is damaged (Fc molecules desorbing) or a bubble is intermittently blocking part of the electrode
Signal below LOD: $A_{\mathrm{peak}} < 3\sigma_{\mathrm{noise}}/\sqrt{N_{\mathrm{pts}}}$	(triggered)	Not triggered	Probe coverage is too low to detect, or signal is lost through complete cleavage; Layer 13 reports $A_{\mathrm{area}} = 0 \pm A_{\mathrm{LOD}}$

Table 1: Automated quality diagnostics applied to every SWV peak extraction. ν is the number of degrees of freedom in the Gaussian fit ($N_{\mathrm{pts}} - 3$, where N_{pts} is the number of points in the peak window); σ_{noise} is the per-point noise from Layer 14.

VII. Edge Cases

Double MB peak

If two distinct MB populations exist on the electrode (e.g., intact probe with MB at the full tether distance, and a cleaved stub with MB at a shorter distance), the SWV voltammogram may show two partially overlapping MB peaks at slightly different formal potentials. The single-Gaussian fit would return a large σ_E and a high χ^2/ν. The diagnostic table (Section VI) flags this case, and the pipeline switches to a two-Gaussian fit within the MB window. The two fitted areas are reported separately; Layer 13 uses only the intact-probe peak (the peak at $E_{\mathrm{peak}} \approx E_{\mathrm{MB}}^{\circ\prime}(\mathrm{pH})$).

Signal clipping

If $A_{\mathrm{peak}} > 0.9 \times I_{\mathrm{range,TIA}}$, the transimpedance amplifier has saturated and the peak top is clipped. The fitted Gaussian amplitude is unreliable. This flag triggers a recommendation to reduce the probe coverage Γ_0 or the TIA gain.

Signal below detection limit

If the peak amplitude is not statistically distinguishable from the noise (see Table 1, last row), Layer 13 is not called for this channel. The result is reported as a censored measurement: $A_{\mathrm{area}} < A_{\mathrm{LOD}}$.

VIII. Frequency Diagnostic Implementation

From Layer 11 (Section XI), the slope of $\log A_{\mathrm{area}}$ versus $\log f_{\mathrm{SWV}}$ distinguishes surface-confined from diffusion-controlled redox. This diagnostic is run once per electrode batch (not on every cartridge):

$$\frac{d\log A_{\mathrm{area}}}{d\log f_{\mathrm{SWV}}} = \begin{cases} 1 & \text{reversible surface-confined } (f \ll k^0/2) \\ 0\text{–}1 & \text{quasi-reversible surface-confined} \\ 0.5 & \text{diffusion-controlled (dissolved MB)} \end{cases} \tag{VIII.1}$$

The area A_{area} is extracted by the Gaussian fit at each frequency using the same pipeline described in this layer. The slope of the log-log plot confirms which regime the electrode is operating in and validates the Layer 11 model.

IX. Layer Outputs

Output	Physical meaning	Typical value	Uncertainty	Feeds
$A_{\text{area}}^{\text{MB}}$	Integrated MB peak area	0.05–5 nA V	$\sigma_{A_{\text{area}}}$ (Eq. IV.3)	Layer 13
$A_{\text{area}}^{\text{Fc}}$	Integrated Fc peak area	0.1–10 nA V	$\sigma_{A_{\text{area}}}$	Layer 13
$E_{\text{peak}}^{\text{MB}}$	Measured MB formal potential	-0.29 ± 0.06 V vs on-chip RE	$\sigma_{E_{\text{peak}}}$	Layer 10 cross-check
σ_E^{MB}	MB peak width	≈ 20 mV	σ_{σ_E}	Quality diagnostic
Quality flags	Boolean flags from Table 1	All clear	—	Data pipeline

Table 2: Outputs of Layer 12 passed to downstream layers. All uncertainties are at the 1σ level and come from the nonlinear least-squares covariance matrix (Eq. IV.3).

X. Limitations

1. **The Gaussian is an approximation to sech^2.** The systematic error in the fitted area from the sech^2/Gaussian discrepancy is $\approx 0.3\%$ (Section I). This is negligible compared to other uncertainties, but the numerical implementation in `peak_extraction.py` can optionally use the exact sech^2 model for verification.

2. **The single-Gaussian fit assumes one probe population.** If the SAM contains probes with a broad distribution of $E_{\text{MB}}^{\circ\prime}$ values (e.g., from heterogeneous PEG linker lengths), the true peak is a superposition of Gaussians at different centres. The fitted single Gaussian overestimates σ_E and produces a biased area. The frequency diagnostic (Section VIII) can distinguish this from quasi-reversible broadening: kinetic broadening increases with f, whereas heterogeneous broadening is frequency-independent.

3. **The cubic baseline polynomial fails near the O_2 reduction wave.** If the cathodic scan limit is close to the O_2 wave (at approximately -0.5 to -0.6 V vs Ag/AgCl in aerated buffer), the wave tail introduces a nonlinear feature at the MB baseline anchor region. In this case, a spline through more anchor points, or a separate exponential term for the O_2 contribution, must be added to Eq. (III.3).

4. **The Fc peak width prediction assumes $n_e = 1$ and the reversible limit.** If Fc is at a distance where $k_{\text{ET,Fc}}^0$ is not much larger than $2f_{\text{SWV}}$, the Fc peak will broaden and its area will drop. Section IV applies the same area-over-height argument to Fc as to MB, so the area is still the preferred metric; but if the Fc peak broadens enough to overlap with the MB peak tail, the two-Gaussian simultaneous fit (Section V) is required.

References

All papers in `electrochemical_device/papers/layer_12/`.

[1] Mirceski V, Skrzypek S, Stojanov L (2018). Square-wave voltammetry: a review on the recent progress. *ChemTexts* 4:17. DOI: 10.1007/s40828-018-0073-0. **mirceski2018.pdf** – surface-confined reversible SWV peak shape (sech^2 form), FWHM, quasi-reversible broadening, and peak area behaviour confirmed.

[2] Savitzky A, Golay MJE (1964). Smoothing and differentiation of data by simplified least squares procedures. *Anal Chem* 36:1627–1639. DOI: 10.1021/ac60214a047. **savitzky1964.pdf** – convolution weights $\{c_k\}$ for polynomial smoothing (Table 1), noise reduction factor, and peak-preservation property confirmed.

[3] Baker BR, Lai RY, Wood MS, Doctor EH, Heeger AJ, Plaxco KW (2006). An electronic, aptamer-based small-molecule sensor for the rapid, label-free detection of cocaine in adulterated samples and biological fluids. *J Am Chem Soc* 128:3138–3139. DOI: 10.1021/ja056957p. **baker2006.pdf** – SWV scan range and 2 mV step size confirmed; on-chip RE peak position reference confirmed.

[4] Lubin AA, Lai RY, Baker BR, Heeger AJ, Plaxco KW (2010). Electrochemical, structure-switching biosensors: quantitative analysis of the usable dynamic range. *Anal Chem* 78:5671–5677. DOI: 10.1021/ac0601417. **lubin2010.pdf** – MB peak position, probe density range, and per-scan signal stability confirmed.

[5] White RJ, Plaxco KW (2010). Exploiting binding-induced changes in probe flexibility for the optimization of electrochemical biosensors. *Anal Chem* 82:73–76. DOI: 10.1021/ac9028595. **white2010.pdf** – SWV frequency dependence of signal area; peak area robustness to frequency change compared to peak height confirmed.